

# IAQF Submission

## 1 Introduction

Over the past several years, the Magnificent Seven (Mag7) stocks—AAPL, MSFT, AMZN, NVDA, TSLA, META, and GOOGL—have experienced dramatic increases in market capitalization. Together, they account for a historically large share of major indices such as the S&P 500 (Figure 1) and NASDAQ 100. This level of concentration is unprecedented in modern financial history.

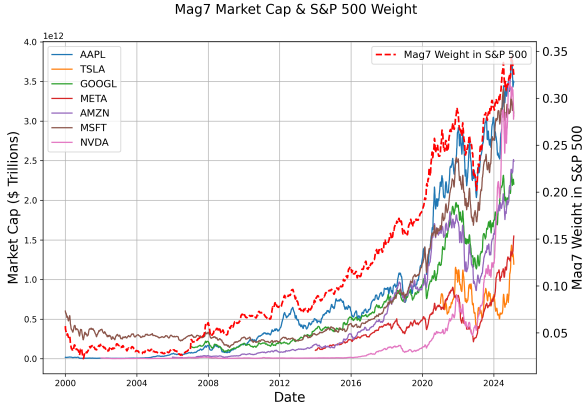


Figure 1: Mag7 Market Cap.

These indices serve multiple functions: asset managers and pensions allocate capital to them; they act as benchmarks for returns and risk exposure; and they serve as proxies for the overall market in hedging and estimation exercises. The dominance of the Mag7 stocks is a relatively new phenomenon and remains understudied. This paper examines its effects across multiple dimensions.

Our research uses daily data on stock prices, at-the-money options, and market capitalization for all S&P 500 constituents since 2000, with annual reconstitution. Additionally, we included SPY options data over the same period. In the final section, the dataset is expanded to cover over 10,000 stocks, incorporating historical data dating back to 1930.

We explore three major implications of the Mag7's dominance. First, we examine its influence on beta—decomposing beta to study the impact of these stocks, and what this means for using beta for investment decisions. Second, we analyze the effects on option pricing and volatility, investigating how the Mag7 influences index distributions and the pricing of index options. Fi-

nally, we assess the implications for trading strategies, particularly long-small-cap, short-large-cap approaches, and how these findings can inform strategy construction.

Our paper provides insights into both the broad market effects of the Mag7 and its differential impact across industries. Our work helps explain the results seen in the last several years, not just in terms of returns but also through the structure of correlation between stocks. We shed light on potential implications of this for future realizations of risk. We also demonstrate how an understanding of the dynamics which drive the performance of a subset of stocks like the Mag7 can be used to modify a common trading strategy like the size factor without the unrealistic need to avoid shorting the best performers.

## 2 Beta

### 2.1 Mag7's Marginal Beta Contribution

To explore the implications of Mag7 concentration, we decompose the market beta of each stock into two components, which we call  $\beta^*$ . Each  $\beta^*$  reflects the influence of either the Mag7 or the remaining 493 stocks:

$$\beta_{SP500} = \beta_{M7}^* + \beta_{493}^* \quad (1)$$

$$\beta_{M7}^* = wt_{M7} \times \frac{\text{var}(R_{M7})}{\text{var}(R_{SP500})} \times \beta_{M7} \quad (2)$$

$$\beta_{493}^* = wt_{493} \times \frac{\text{var}(R_{493})}{\text{var}(R_{SP500})} \times \beta_{493} \quad (3)$$

where:

- $wt_{M7}$  and  $wt_{493}$  are the total market cap weights of Mag7 stocks and the rest of the S&P 500 stocks.
- $R_{M7}$  and  $R_{493}$  are the market cap-weighted index returns.
- $\beta_{SP500}$ ,  $\beta_{M7}$ , and  $\beta_{493}$  are the rolling 1-year betas of each stock with respect to these three index returns.

Equation 2 and 3 further decomposes each  $\beta^*$ , allowing us to see how the Mag7 contributes to a stock's market beta. There are two channels of influence: (1) increasing weight ( $wt_{M7}$ ) of Mag7 and (2) Mag7's economic co-movement effect ( $\beta_{M7}$ ).

Figure 2 shows the average contribution of  $\beta_{M7}^*$  and  $\beta_{493}^*$  to total beta. Mag7's contribution has shown a steady increase since 2000, and two significant increases can also be observed in 2020 and 2022, corresponding to their rapid market cap increases seen in Figure 1.

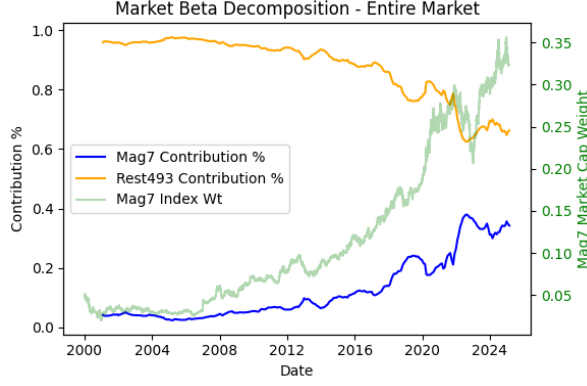
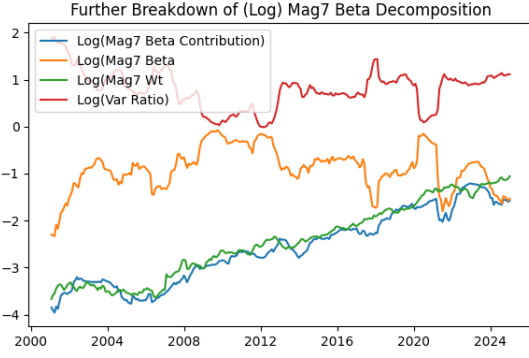
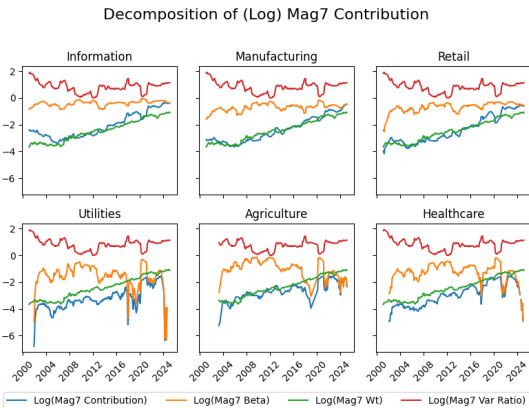


Figure 2: Market Beta Decomposition

## 2.2 Log Decomposition

Figure 3: Breakdown of  $\text{Log}(\beta_{M7}^*)$ 

To further assess the relative importance of the weight and co-movement channels, we do a log transformation of Eq. 2 to make the relationship additive. As shown in Figure 3, the fact that  $\text{Log}(\beta_{M7}^*)$  moves very closely with  $\text{Log}(wt_{M7})$  demonstrates that the weight factor is the primary driver of the upward trend in Mag7's beta contribution, while the economic co-movement factor is of second-order significance.

Figure 4: Industry-level Breakdown  $\text{Log}(\beta_{M7}^*)$ 

However, this market-average view masks important insights which become evident once we group stocks into industries. We show these results in Figure 4. For this analysis, we calculated the market cap weighted industry index returns and performed the Log decomposition for selected industries.

Since the Mag7 weight remains consistent across industries, a relative drop in  $\beta_{M7}^*$  within an industry primarily results from a decline in the industry's own  $\beta_{M7}$ .

On the first row we show industries that include or have a close economic connection to Mag7. They exhibit relatively stable  $\beta_{M7}$ , independent of changes in the industries'  $\beta_{M7}^*$ . This stability is due to the correlation of these industries with Mag7 remaining high despite changes in Mag7 returns (i.e., these industries closely follow Mag7).

However, a different pattern emerges in industries that are economically distant from the Mag7, as shown in the second row. There we see much greater variation in  $\beta_{M7}^*$  and this is driven by large swings in the industries'  $\beta_{M7}$ 's. This is particularly noticeable when  $\beta_{M7}$  experienced sharp falls while Mag7 weights are increasing fast. The common factor that was driving Mag7 out-performance was not (yet) reaching these industries and this caused their  $\beta_{M7}$  to fall, which in turn drove changes in  $\beta_{M7}^*$  and overall market beta. However, the  $\beta_{M7}^*$  don't appear to stay permanently lower, they rebound - at least partially - once the strong Mag7 outperformance period dissipates.

We interpret this important finding as a difference in the “**propagation speed**”. The common factor driving Mag7's rapid increase in market cap is also present in industries like Tech, Manufacturing and Retail, which is reflected in their stable  $\beta_{M7}$ 's. Meanwhile, the factor takes time to impact industries less connected to the Mag7, leading to a fall in their  $\beta_{M7}$  and then a lagged catch up as presumably some of fundamental drivers of the Mag7 propagate through the economy and reach other firms.

## 2.3 Beta Deviation

In this section we present another way to view the differential propagation speed of the factors driving the Mag7 across industries. In section 4 we use this idea of differential propagation speeds to inform our small-cap trading strategy.  $\beta_{493}^*$  is a measure of the impact that non-Mag7 firms have on a stock's market beta. The ratio of  $\beta_{493}^*/\beta_{SP500}$  tells us how much of a wedge the Mag7 has driven between a stock's  $\beta_{SP500}$  and its beta to the remaining 493 firms.

Assuming S&P500 is a diversified stock portfolio, the increasing impact of Mag7 has introduced a concentration risk that reduces the index's diversification. By defining **Beta Deviation** as  $\beta_{493}^*/\beta_{SP500}$ , the remaining 493 stocks are used as a proxy for the *true*  $\beta_{SP500}$  free from the effects of Mag7 concentration.

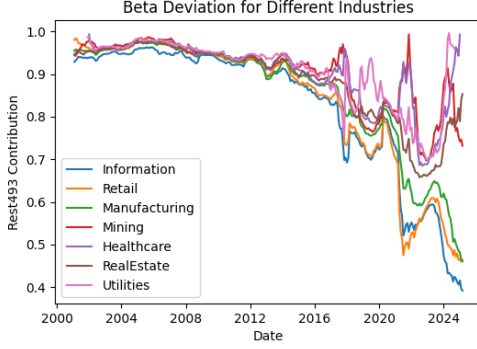


Figure 5: Industry-level Beta Deviation

When compared across industries, significant deviations are observed. During 2016-2018, 2020-2022, and 2024 to the present, the Mag7 drove significant deviations in beta with the non-Mag7 stocks, with the first two periods eventually converging. This pattern aligns with earlier observations indicating differences in Mag7’s propagation speed across industries.

- The 2016-2018 phase saw the first boom of Mag7, and industries differ on their response to the rally.
- The 2021 divergence can be attributed to Fed tightening and a technology selloff, where defensive sectors shifted away from Mag7 exposure, increasing the impact of the 493.
- The current divergence since 2024 can mainly be summarized to the AI hype. From an outlook perspective, using history as reference, this may indicate that the dispersion will close again, and the currently low correlation among industries will increase.

## 2.4 Covariance Prediction

Previously, we examined the spillover effect and the propagation of Mag7 price movements to other industries. Now, we turn our attention to how Mag7 price fluctuations influence the beta of the entire market. This analysis provides insights into the market’s systematic risk and the degree of market cohesion in response to changes in Mag7 prices, offering practical implications for investment strategies. For example, if we can anticipate the price change of a Mag7 stock in the coming month, we could predict the resulting change in market covariance. Or, absent a prediction, understanding this relationship allows for informed scenario analysis on market-level risk given different possible changes in Mag7 returns. Both use cases could enhance risk management.

To estimate this effect, the covariance change of an industry with a Mag7 stock in month  $t$  is regressed on the cumulative return of that stock in the same month, controlling for size effects by including an absolute value term of

the return. The regression formula is as follows:

$$\Delta_{t+1} \text{Cov}(r_{\text{industry}}, r_{\text{mag7\_stock}}) = c + \beta_0 r_{\text{mag7\_stock}, t+1} + \beta_1 |r_{\text{mag7\_stock}, t+1}| + \epsilon \quad (4)$$

Where  $\Delta_{t+1} \text{Cov}$  is the change of covariance from  $t$  to  $t+1$ , and  $r_{\text{mag7\_stock}, t+1}$  is the cumulative return from  $t$  to  $t+1$ . The model is trained universally by pooling data from all industries for each Mag7 stock using an expanding window starting from 2000, with exponential weighting on observations. The model is retrained monthly to predict the following month’s covariance changes.

The forecast is compared with a naive forecast assuming a zero covariance change. The MSEs of the naive and forecasted ( $\Delta_{t+1} \widehat{\text{Cov}}$ ) predictions are calculated as follows.

$$\begin{aligned} MSE_{\text{naive}} &= \text{mean}(\Delta_{t+1} \text{Cov}(r_{\text{industry}}, r_{\text{mag7\_stock}})^2) \\ MSE_{\text{forecast}} &= \text{mean}((\Delta_{t+1} \text{Cov}(r_{\text{industry}}, r_{\text{mag7\_stock}}) - \Delta_{t+1} \widehat{\text{Cov}})^2) \end{aligned} \quad (5)$$

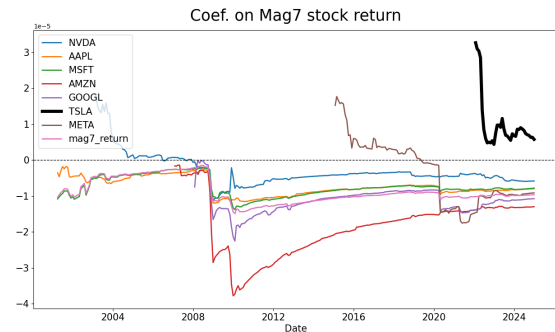
The MSE comparison shows a significant average decrease of 6.6% for all Mag7 stocks, with the most substantial improvement for Tesla at 80%.

| Stock        | NVDA  | AAPL | MSFT | AMZN |
|--------------|-------|------|------|------|
| MSE Drop (%) | 14.3  | 3.3  | 2.6  | 6.8  |
| Stock        | GOOGL | TSLA | META | Mag7 |
| MSE Drop (%) | 5.7   | 80.0 | 12.5 | 6.6  |

Table 1: Forecast improvements

Furthermore, the key market spillover effect can be analyzed through the model parameters.

The negative  $\beta_0$  coefficients (Figure 6) indicate that when Mag7 stocks decline, industry correlations with Mag7 increase. This is likely due to common factors that affect both Mag7 and other industries, leading to synchronized market drawdowns. This aligns with the volatility smirk, where downside risks amplify systematic risks and implied volatility. As market leaders, Mag7 stocks embody a large share of systematic risk, influencing broader market correlations.

Figure 6:  $\beta_0$ : Coefs on Mag 7 stock return

The positive  $\beta_1$  coefficients (Figure 7) reveal that shock magnitude positively correlates with systematic risk, showing that large shocks tend to move markets collectively.

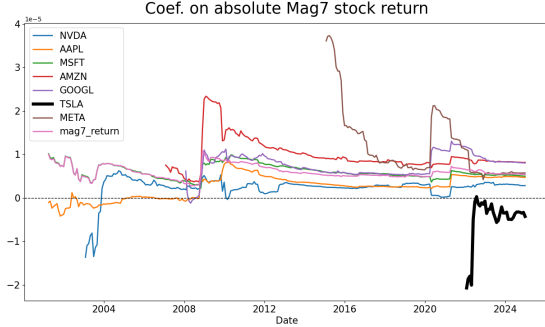


Figure 7:  $\beta_1$ : Coefs on absolute Mag 7 stock return

Finally, Tesla is an outlier, with its coefficient sign the opposite of the other Mag7 firms. Unlike other Mag7 stocks, Tesla's positive shocks increase market correlation more than negative shocks. The reason behind is that Tesla's drops are mainly driven by idiosyncratic factors such as declining sales, increased competition, leadership controversies, and product challenges, while its rises are more aligned with systematic market trends. We will conduct a case study in the next section to illustrate.

The analysis is reinforced by observations post-2020, where Mag7 drawdowns typically coincided with volatility spikes. Conversely, Mag7 growth phases generally led to reduced systematic risk and market volatility.

## 2.5 Case Study: TSLA

TSLA's drop from second half of 2023 to first half of 2024 represents an example of our findings. During this period, TSLA dropped by 43%, which led Mag7 as a whole to only moderately increase 6% during this period, despite NVDA's 60% jump and META's 34% increase.

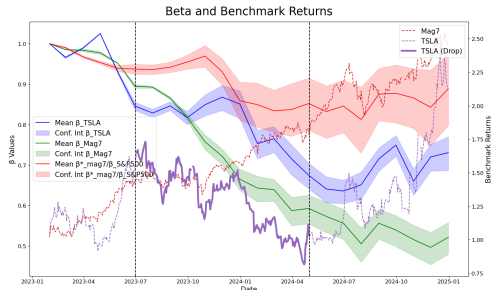


Figure 8: Influence of Tesla's drop on betas

During Tesla's decline, the covariance between other industries and Tesla diminishes as evidenced by the decrease in Tesla's beta (blue line). This is because the decline is driven by idiosyncratic factors such as declining

vehicle sales (45 percent drop in Europe), increased competition from Chinese EV makers, CEO Elon Musk's polarizing political involvement, delays in Full Self-Driving technology, and labor disputes in Sweden.

Simultaneously, with most other Mag7 stocks performing well, the covariance with other industries also falls. This dual effect causes the Mag7 beta (green line) to decline more sharply, with the rate of decline moderating only after Tesla's drop stabilizes. The rapid decrease in Mag7 beta reduces the Mag7 decomposition percentage (red line), despite high market concentration. This shift indicates a more diversified market environment with lower systematic risk.

In summary, Tesla's price movements can amplify beta fluctuations, particularly when moving against the broader market trend. In this case, Tesla's decline drives a significant reduction in beta, suggesting a market profile with increased idiosyncratic risk.

## 3 Options

### 3.1 Index Volatility

Understanding the impact of Mag7 stocks on the S&P 500 requires analyzing their influence on the options market, particularly how they shape the distribution of index returns. Since the index return series is inherently dependent on the weights of its constituents, we examine the equation for index volatility:

$$\sigma_{\text{index}}^2 = \sum_{i=1}^N w_i^2 \sigma_i^2 + 2 \sum_{i=1}^N \sum_{j=i+1}^N w_i w_j \rho_{i,j} \sigma_i \sigma_j \quad (6)$$

Given an increase in concentration, assuming all other variables remain constant, we would expect an increase in index volatility due to the non-linearity of the equation. However, this pattern is not observed in the data. Our analysis shows that implied volatility does not exhibit a significant increase during the period of heightened concentration (2020–2024), despite the surge in the market capitalization of the Mag7.

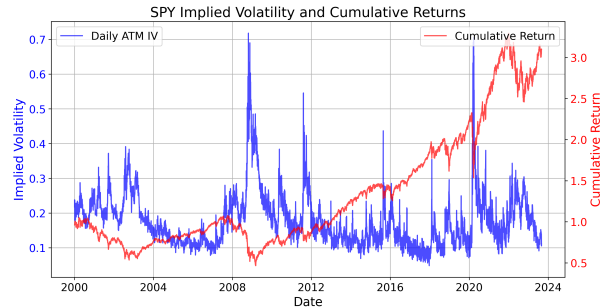


Figure 9: ATM implied volatility of S&P500 with returns.

Observing the changes since 2020, we identify distinct



periods of declining index volatility from 2020–2022 and 2022–2024. This directly contradicts the expected outcome of increased index concentration. However, considering the broader market conditions during this period, these results are not surprising. Figure 9 clearly shows that market returns have an inverse relationship with volatility.

It is natural for periods of increased concentration to coincide with market rallies. As a constituent's price outperforms the broader market, its weight in the index rises. Given the empirical evidence, other factors in the index volatility equation adjust to offset the increase in concentration. This offset could stem from changes in either the correlation structure or individual stock volatilities. To measure this, we estimated both the realized correlation matrix and the option-implied correlation matrices.

To estimate the realized correlation, we used Principal Component Analysis (PCA) to reduce the dimensionality of the problem. At each time step, we computed a new correlation matrix for all 500 index constituents using one year of stock returns. Given the large number of parameters to estimate, we first reduced the return matrix using the top 25 principal components, reconstructing the correlation matrix:

$$C = V\Lambda V^T \quad (7)$$

$$C_k = V_k \Lambda_k V_k^T \quad (8)$$

- $C$ :  $N \times N$  correlation matrix
- $V$ :  $N \times N$  matrix of eigenvectors
- $\Lambda$ :  $N \times N$  diagonal matrix of eigenvalues
- $k$ : number of principal components

Since realized returns are measured under the physical measure, we applied an adjustment to convert them to the risk-neutral measure. We used the method introduced by Busse and Volkov (2012) in *Measuring Equity Risk with Option-Implied Correlations (Review of Financial Studies, 25(10), pp. 3113–3140)*.

$$\rho_{ij}^Q = \rho_{ij}^P - \alpha(1 - \rho_{ij}^P) \quad (9)$$

- $\rho_{ij}^Q$ : implied correlation
- $\rho_{ij}^P$ : realized correlation
- $\alpha$ : risk-adjustment parameter

We solve for the alpha factor by matching the observed index variance from options with the implied variance derived from individual constituents' option prices.

$$\alpha = \frac{\sigma_{M,Q}^2 - \sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{i,Q} \sigma_{j,Q} \rho_{ij}^P}{\sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{i,Q} \sigma_{j,Q} (1 - \rho_{ij}^P)} \quad (10)$$

- $\sigma_{M,Q}^2$ : index risk-neutral variance
- $\sigma_{i,Q}$ : implied volatility of stock  $i$
- $w_i$ : weight of stock  $i$  in the index
- $\rho_{ij}^P$ : historical correlation

Using this methodology, we estimate an implied correlation matrix at each time step. By examining the top principal components (PCs) of the correlation matrix, we gain insights into the factors driving market-wide correlation structures. Intuitively, the first PC captures the 'market factor,' while the second and third load on factors that deviate from market-wide movements.

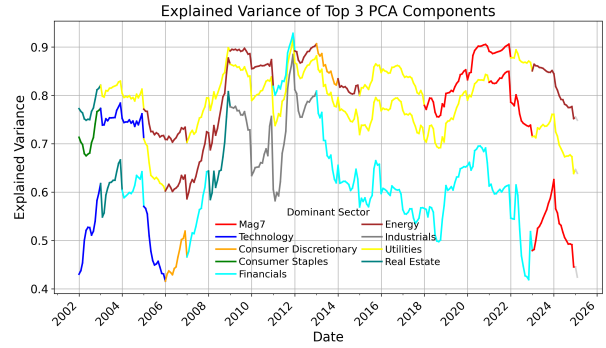


Figure 10: Explained variance of correlation

In Figure 10, we plot the explained variance of the top three PCs. At each time step, we identify the 'dominant' sector in each PC by averaging the loadings of stocks within each sector and selecting the one with the highest component. The sectors include GICS sectors, along with the Mag7 grouping. As Mag7 stocks increase in concentration, their influence on the correlation matrix grows, as evidenced by Mag7 dominance shifting from the third PC to the second, and eventually to the first PC between 2018 and 2024. This shift reflects the increasing dominance of Mag7 stocks in the correlation structure.

Additionally, we observe a decreasing trend in the explained variance of the first PC, the 'market factor,' while the overall explained variance of the top three PCs remains relatively stable. This suggests that the correlation structure is becoming less driven by the overall market factor and more influenced by deviations from it. We can further analyze this effect by examining individual sectors. Sectors closely tied to Mag7 stocks should exhibit stable or increased correlations with the market, while those less related should show a decline in correlation.

In Figure 11, we plot the realized and implied correlations of different industries with the S&P 500 over time, illustrating the impact of the Mag7 on the market's correlation structure. Industries such as consumer staples, energy, and healthcare, which have low correlation with the Mag7 stocks, have seen a decline in correlation with the overall index. In contrast, industries like consumer discre-

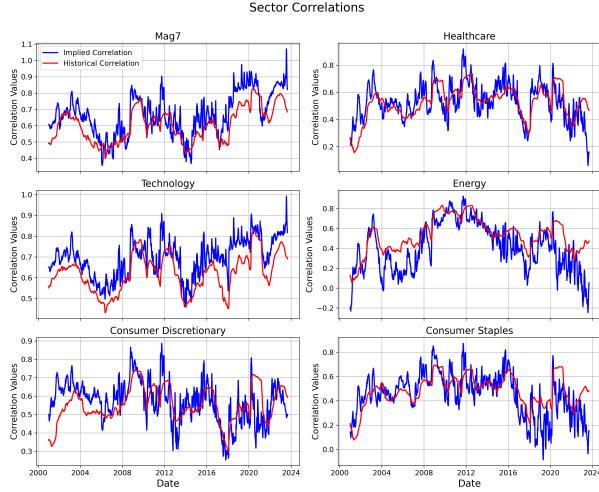


Figure 11: Sector correlations

tionary and technology, which are more closely tied to the Mag7 stocks, have experienced an increase in correlation with the index. This trend is even more pronounced in the options-implied correlation, suggesting that this forward-looking measure is pricing in a more extreme divergence in correlations across industries.

The reduction in correlation offsets the increase in concentration, stabilizing overall index volatility. This decline in correlation is a characteristic of the current bull market. However, if market growth slows or reverses, correlations may revert to previous levels without a corresponding decrease in concentration. If concentration remains high or continues to rise, the potential for increased market volatility during the next economic downturn becomes even greater.

### 3.2 Higher Order Moments

Another way we assess the impact of increased concentration on options markets is by examining higher-order moments.

As an index becomes more concentrated, a few large stocks carry disproportionate weight, leading to larger, asymmetric movements in index returns. Due to this, we expect greater concentration to increase skewness—reflecting heightened demand for put options to hedge idiosyncratic risks—and kurtosis, as returns exhibit fatter tails compared to a more diversified index.

We use the Herfindahl-Hirschman Index (HHI) as a proxy for market concentration, while call-put skew and the butterfly spread serve as proxies for the skewness and convexity of the SPX options' implied volatility structure. We compute these metrics using the following formulas:

$$\text{HHI Index} = \sum_i w_i^2 \text{ for } i = 1, 2, \dots, n \quad (11)$$

- $w_i$ : market share of each stock

$$\text{Skew} = \frac{\sigma_{90\% \text{ put}} - \sigma_{110\% \text{ Call}}}{\sigma_{100\%}} \quad (12)$$

$$\text{Convexity} = \frac{\sigma_{90\% \text{ put}} - 2 \times \sigma_{100\%} + \sigma_{110\% \text{ Call}}}{\sigma_{100\%}} \quad (13)$$

Figure 12 shows that as market concentration increased and fluctuated after 2020, both skewness and convexity exhibited comovements with the HHI index. The visual evidence suggests that after the HHI reached an all-time high in 2020, skew and convexity began to exhibit a stronger correlation with market concentration. This relationship is further illustrated in the box plots, which depict the interaction between these variables from 2020 onward. The upward trend indicates that higher market concentration is associated with greater skewness and convexity, reinforcing the idea that increased concentration heightens return asymmetry and amplifies tail risk.

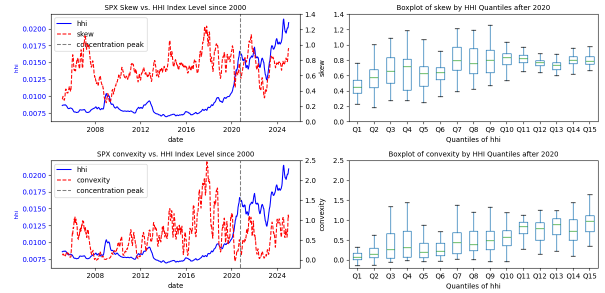


Figure 12: SPX Skew and Convexity vs. HHI Index

### 3.3 Index Option Sensitivity

Another consequence of the surge in market capitalization of the Mag7 stocks relative to the index is the increased impact of their movements on the overall index. In the context of options, we consider this influence as an option sensitivity to movements in individual constituents, particularly the Mag7 stocks. We quantify this effect through a "Mag7 Greek," which captures their contribution to index option risk.

$$\frac{\partial C}{\partial \sigma_i} = \text{Vega} \times \frac{\partial \sigma_{\text{index}}}{\partial \sigma_i} \quad (14)$$

$$\frac{\partial \sigma_{\text{index}}}{\partial \sigma_i} = \frac{w_i}{\sigma_{\text{index}}} \sum_j w_j \rho_{i,j} \sigma_j \quad (15)$$

Equations 15 and 16 together define the sensitivity of an index option to an individual stock's volatility. We define the Mag7 Greek as the sum of the individual contributions of each Mag7 stock. It is most relevant to consider the Mag7 Greek as a percentage of overall SPX vega, as this eliminates dependence on the underlying index price. Notably, the Mag7 Greek is highly influenced by index

weight, making it expected to move closely with it. Any deviations from this trend can be attributed to other factors in the equation—such as correlation, and the fact that as the Mag7 weight increases, the weights of other constituents decrease.

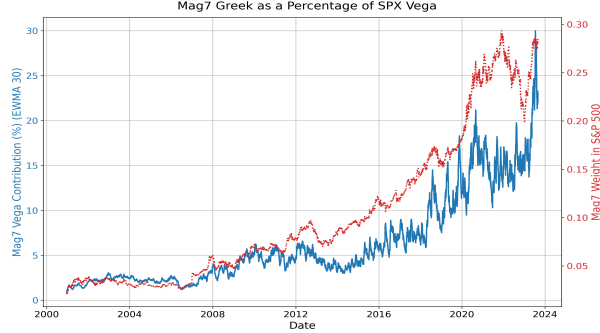


Figure 13: Mag7 Greek sensitivity percentage and Mag7 weight.

Figure 13 displays the Mag7 Greek as a percentage of SPY vega and Mag7 weight in the index. This reinforces the impact of sparse market correlations. The deviation of the Mag7 Greek from the Mag7 weight can be attributed to a fragmented overall market correlation structure during this period. The most pronounced deviation occurs between 2020 and 2024, aligning with our earlier findings.

The Mag7 Greek quantifies the impact of Mag7 stocks on both index price and volatility. In the next section, we examine how major volatility shocks in Mag7 stocks propagate to the broader market through a case study approach.

### 3.4 Empirical Analysis: Does Mag7 Volatility Surge Impact the Market

We conduct an empirical analysis of historical volatility in Mag7 stocks and SPX options. We construct a market cap-weighted index for Mag7 stocks and compute the 30-day annualized volatility. During the first two years of the concentration peak period (2020–2022), a volatility spike occurred due to the COVID-19 shock. However, after the initial recovery, both Mag7 and market volatility trended downward. After October 2021, volatility began rising again, yet spikes in Mag7 volatility were not always accompanied by simultaneous increases in SPX implied volatility, despite the market concentration level remaining high. We attribute this to the declining correlation between Mag7 stocks and the broader market.

Table 2 above highlights certain periods when Mag7 volatility spiked, yet SPX ATM implied vol did not always experience an equivalent increase. This divergence suggests that volatility shocks within Mag 7 stocks do not necessarily propagate to broader market volatility, particularly after 2022.

| Start Date | End Date   | $\% \Delta \sigma_{Mag7}$ | $\% \Delta \sigma_{SPX}$ |
|------------|------------|---------------------------|--------------------------|
| 2022-04-25 | 2022-06-01 | +43%                      | +0.01%                   |
| 2022-09-10 | 2022-11-15 | +73%                      | +0.07%                   |
| 2024-01-30 | 2024-03-15 | +64%                      | +16%                     |
| 2024-06-25 | 2024-08-10 | +40%                      | +47%                     |

Table 2: Mag7 volatility and SPX implied volatility do not move closely after 2022

## 4 Long Small-Cap and Short-Large Cap Strategy Analysis

After analyzing the effects of concentration within the S&P 500, we now extend our analysis to the entire market. Specifically, we examine how increasing market concentration affects a long small-cap/short large-cap strategy—and how we might improve this strategy by incorporating our insights on information propagation and lead-lag effects. Our universe is every stock in the CRSP database, updated monthly. This gives us a sample of around 2,000 stocks at the onset, which rises to around 10,000 by the end of 2024.

### 4.1 Implications of Concentration

We begin by building a benchmark strategy. To do this, we classify the top 33% of stocks by market capitalization as large caps and the bottom 33% as small caps, using an expanded universe of stocks. Within each industry, we construct a portfolio that goes long an equally-weighted basket of small-caps and short an equally-weighted basket of large-caps. Each of these industry portfolios receives equal weight in the overall portfolio. The result is a portfolio that is both dollar- and industry-neutral with capital distributed evenly across all industries. The cumulative return plot (Figure 14) clearly indicates that while the strategy performed reasonably well over the past 50 years, its performance has deteriorated recently. This underperformance suggests that increased market concentration—where a few large-cap stocks dominate—undermines the traditional long small-cap/short large-cap approach.

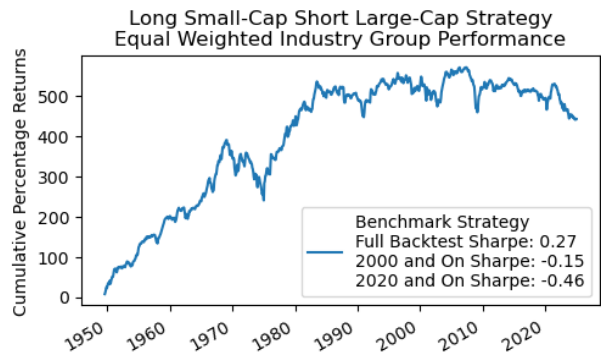


Figure 14: Benchmark Portfolio Performance

This observation aligns with our findings in Section 2, where we demonstrated that market capitalization is a more critical driver of market dynamics than individual stock beta. In other words, the overwhelming influence of a few large-cap stocks (such as the Mag7) tends to skew overall market behavior, thereby distorting the performance of strategies that rely on relative size dispersion. We also note that this holds true for the small-cap performance in the industry most influenced by the Mag7, technology.

## 4.2 Strategy Development and Rationale

Building on our beta analysis—which showed that the common factor behind the Mag7’s outperformance propagates at different speeds across industries—we design a trading strategy to improve portfolio performance. Our approach incorporates two key indicators: **Dispersion** and **Influence**.

### 4.2.1 Dispersion Indicator

The Dispersion indicator is designed to capture the idea that when there is a large performance gap between large-cap and small-cap stocks within an industry, small-cap stocks are likely to catch up. Specifically:

- Within each industry, we calculate the change in the (log) market capitalization of the top third of companies relative to that of the bottom third over a trailing three-month period.
- To avoid further penalizing industries where small caps are growing faster than large caps, the dispersion measure is clipped at a minimum value of zero.
- The resulting percentage change is then smoothed using an exponentially weighted moving average (EWMA) with a 12-month half-life.

This indicator reflects the intuition that a large dispersion signals a lag in the price adjustment of small-cap stocks. When dispersion is high, it implies that large-cap stocks have already reacted to prevailing information, while small caps are delayed in their response—setting the stage for a catch-up or momentum effect.

### 4.2.2 Influence Indicator

The Influence indicator is designed to identify the “hot” or leading industry group and to capture how other parts of the market follow its momentum. The logic here is twofold:

- **Percentage Change in Market Cap:** For each industry group, we compute the percentage change in total market capitalization over a three-month window:

$$\Delta S_i(t) = \frac{S_i(t) - S_i(t-3)}{S_i(t-3)},$$

where  $S_i(t)$  denotes the market capitalization of industry  $i$  at time  $t$ .

- **Identifying the Hottest Industry Group:** The “hottest” industry group at time  $t$  is defined as the one with the maximum normalized percentage change:

$$H(t) = \arg \max_i \Delta \tilde{S}_i(t).$$

Because investment decisions are made based on historical data, we use the previous period’s hot industry group (i.e.,  $H(t) = H(t-1)$ ).

- **Correlation to the Hot Industry Group:** Next, we calculate an exponentially weighted correlation matrix for the returns of large-cap industries (using a 60-month half-life). For industry  $i$  at time  $t$ , let  $R_i(t)$  be its return and  $\mu_i$  its exponentially weighted mean. The correlation between industries  $i$  and  $j$  is given by:

$$\text{Corr}_{i,j}(t) = \frac{\text{EWM}_\lambda[(R_i - \mu_i)(R_j - \mu_j)]}{\sqrt{\text{EWM}_\lambda[(R_i - \mu_i)^2] \cdot \text{EWM}_\lambda[(R_j - \mu_j)^2]}}.$$

Industries with higher positive correlations to the hot industry are expected to eventually benefit from its influence. Thus, we tilt our small-cap allocation toward these industries.

By leaving the large-cap positions unchanged, our strategy aims to capture the natural flow of influence from the initial responders (large caps) to the lagging small caps. This is important because adjusting the large-cap positions might introduce over fitting—any observed out performance from reducing exposure to dominant large caps is likely to be data-mined rather than reflective of a genuine, repeatable phenomenon.

## 4.3 Testing

We evaluate these indicators’ potential to improve upon the benchmark by initially using them as two standalone tilts—each with a standard 1% deviation (netting to zero) applied to the overweights and underweights of the industry groups. Next, we develop a combined strategy where the weight of each indicator is optimized based on its historical performance. Importantly, throughout all tests, we adjust only the composition of the small-cap long positions.

**Single-Indicator Tilts:** The results of the single-indicator tilts are displayed in Figure 15, which plots the cumulative return in excess of the benchmark and shows meaningful improvements. In Table 3, we show the largest overweight tilts over time for each indicator.



Summarizing the information across decades leaves out some detail but it shows that the indicators are identifying small caps in different industries as overweight targets. This suggests that combining them together could provide a diversification benefit.

| 1970's                              | 1980's                              | 1990's                                     |
|-------------------------------------|-------------------------------------|--------------------------------------------|
| <b>Dispersion</b>                   |                                     |                                            |
| Media & Entertainment<br>2.3%       | Telecommunication Services<br>1.2%  | Media & Entertainment<br>1.4%              |
| Semiconductors<br>1.0%              | Banks<br>0.6%                       | Household & Personal Products<br>0.8%      |
| Insurance<br>0.6%                   | Consumer Staples<br>0.5%            | Semiconductors<br>0.6%                     |
| <b>Influence</b>                    |                                     |                                            |
| Capital Goods<br>0.8%               | Capital Goods<br>0.9%               | Capital Goods<br>0.9%                      |
| Consumer Services<br>0.8%           | Financial Services<br>0.7%          | Technology Equipment<br>0.9%               |
| Consumer Durables & Apparel<br>0.7% | Consumer Durables & Apparel<br>0.7% | Consumer Durables & Apparel<br>0.7%        |
| <b>2000's</b>                       |                                     |                                            |
| <b>2010's</b>                       |                                     |                                            |
| <b>2020-2024</b>                    |                                     |                                            |
| REITs<br>1.1%                       | Specialty Retail<br>1.0%            | Commercial & Professional Services<br>1.0% |
| Automobiles & Components<br>0.7%    | Telecommunication Services<br>0.9%  | Automobiles & Components<br>1.0%           |
| Consumer Staples<br>0.7%            | Insurance<br>0.6%                   | Media & Entertainment<br>1.0%              |
| <b>Influence</b>                    |                                     |                                            |
| Capital Goods<br>0.7%               | Capital Goods<br>0.8%               | Financial Services<br>0.7%                 |
| Financial Services<br>0.7%          | Financial Services<br>0.8%          | Software & Services<br>0.7%                |
| Automobiles & Components<br>0.7%    | Media & Entertainment<br>0.6%       | Capital Goods<br>0.7%                      |

Table 3: Highest Up-Weighting of Industry Groups for Dispersion and Influence Tilts by Decade (1970's–2024)

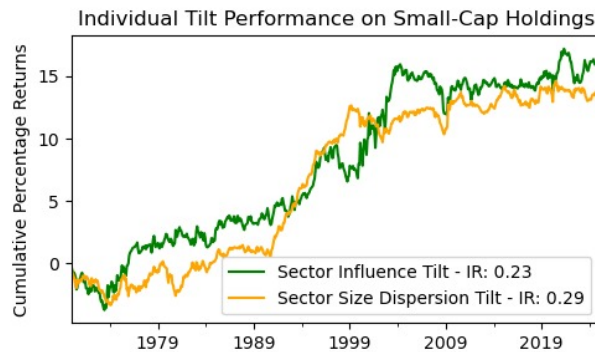


Figure 15: Individual Tilt Performance

**Combining Indicators and Final Optimization:** Finally, we combine the two indicators using an optimizer that assigns monthly weights based on past performance (maximizing the IR with a 5-year EWMA half-life). In addition to the weighted signal, we include a default benchmark position—reflecting the fact that investors are unlikely to persist with a signal if its recent performance is poor, and would have the option to turn both tilts off. The strategy's performance relative to the benchmark is shown in Figure 16 and the evolution of the optimized indicator weights is shown in Figure 17..

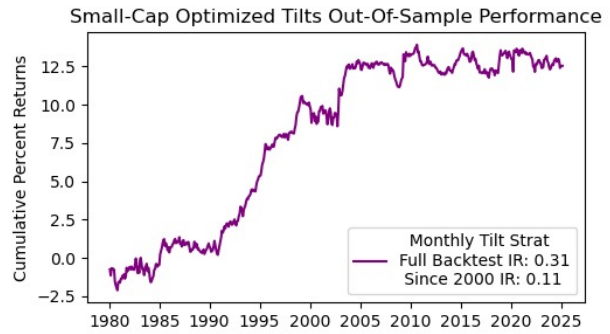


Figure 16: Combined OOS Performance of Tilts

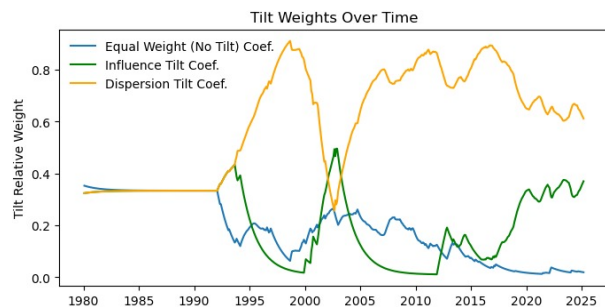


Figure 17: Optimized Tilt Weights

While the Dispersion signal trumps in importance for most of the sample, Figure 17 may reveal valuable insights through the relative changes. There are two explanations for the relative up-weighting of the Influence indicator in the early 2000s and now the 2020s:

- **Influence indicator is working:** An increasing coefficient for the Influence tilt suggests that industry groups closely tied to the leading sector benefit more from that relationship than similar industry groups did in previous periods. Since technology plays a major role in both periods—the post-dot-com reshuffling and the recent rise of the Mag7—this pattern suggests that during times of technological reshuffling, closely related sectors benefit more from innovation and efficiency gains than from other adjacent industries.
- **Dispersion indicator is failing:** Conversely, these two periods could also reflect the Dispersion tilt failing to capture alpha, as large-cap stocks surged without their growth effect trickling down to smaller firms within their industry grouping. This hypothesis is backed by under-performance of the small-cap technology stocks relative to both the overall technology sector and small caps across the market more generally for the last few years.

## 4.4 Final Integrated Logic

In summary, our strategy is driven by the following logical considerations:

- **Benchmark Consistency:** The benchmark, defined as long the bottom third and short the top third of stocks by market cap within each industry, clearly demonstrates that increased concentration undermines performance.
- **Information Propagation:** The Dispersion indicator captures the delayed reaction of small caps relative to large caps, signaling an eventual catch-up when dispersion is high.
- **Industry Influence:** The Influence indicator identifies the leading industry (typically dominated by large caps) and tilts the small-cap allocation toward industries with strong positive correlations to this leader.
- **Avoiding Overfitting:** By not tilting the large-cap side, we avoid potential data-mining biases. This approach focuses on capturing genuine market dynamics—the *propagation of influence from dominant stocks to smaller ones*—rather than merely reflecting the inherent outperformance of large caps.
- **Dynamic Optimization:** The final combination of signals, optimized over a five-year look-back period, allows the strategy to adapt to changing market conditions, thereby improving risk-adjusted returns (as evidenced by the enhanced IR).

This integrated approach not only improves the performance of the long small-cap/short large-cap strategy but also offers a theoretically consistent framework for understanding how market concentration affects relative stock performance.

## 5 Conclusion

This paper examines the implications of the growing dominance of the Magnificent Seven stocks, particularly their impact on beta, options markets, and trading strategies. The dramatic increase in market concentration, driven by the surge of these stocks, presents unique challenges for investors.

Our beta decomposition demonstrated that the Mag7's increasing weight in the S&P 500 is the primary driver of its growing influence on a typical stock's beta. We also showed that the Mag7's influence propagates through the economy at different speeds. We further demonstrated that Mag7 returns impact the correlation structure, and therefore riskiness, of the market. Positive Mag7 returns are associated with widening dispersion in returns to other

stocks, while negative Mag7 returns lead to more synchronized market movements.

Investors need to understand this as it suggests a future fall in Mag7 stocks would not only impact the market directly – since they represent a large weight in the index – but would also increase risk via higher covariances between other stocks. Tesla stands out as an exception, its movements have the opposite effect on covariance, which suggests future shocks to its price are not a source of systematic risk.

Our options market analysis examined how Mag7 concentration impacts index return distributions. Despite increasing concentration, index volatility has remained stable due to a decline in market-wide correlations. However, market concentration has increased skewness and convexity in SPX options, affecting the pricing of out-of-the-money options. This may reflect market recognition of our earlier points that a reversal of Mag7 concentration could lead to more synchronized downside returns across the market.

Lastly, we use the insight that new common factors – like those driving the Mag7 – propagate across industries at different speeds to construct dispersion and influence factors, which in turn are used to modify a size-based long-short strategy. The use of these indicators improves significantly on our benchmark performance and does so without the hindsight bias of simply down weighting shorts in the Mag7.

Taken together, we believe our ideas can be used by managers as a framework for understanding and reacting to the current environment of extreme market concentration.

## References

- Busse, A., & Vilkov, G. (2012). Measuring equity risk with option-implied correlations. *The Review of Financial Studies*, **25**(10), 3113–3140. <https://doi.org/10.1093/rfs/hhs087>.
- Buss, A., Schoenleber, L., & Vilkov, G. (2017). Option-implied correlations, factor models, and market risk. Working paper.
- Kaiser, L. (2017). Portfolio concentration and equity market contagion: Evidence on the “flight to familiarity” across indexing methods. Working paper.
- Pandey, A., & Sharma, A. K. (2023). Effect of index concentration on index volatility and performance. *Asia-Pacific Financial Markets*, **30**, 559–585.
- Invesco. (n.d.). Invesco S&P SmallCap Information Technology ETF (PSCT). Retrieved February 27, 2025, from <https://www.invesco.com/us/financial-products/etfs/product-detail?audienceType=Investor&ticker=PSCT>.